

MEKDELA AMBA UNIVERSITY

College Of Computing and Informatics

Department Of Software Engineering

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ABSTRACT

The computer systems application allows people to quickly and easily browse and manage data in the form of electronics. A web application is any application software that runs in a web browser or is created to enable people to do their day to day activities where ever.

Mekdela amba University does not have its own web based student registration system. Due to that, it could not provide fast and flexible service for its users.

Some of its problem is registration problem, information accessing problem, etc. So there must be a system that can handle these and the like problems. The main purpose of this project was to develop web based registration system for Mekdela amba University to change the manual system of student registration, information announcement, commenting and so on to the web based system by developing dynamic website application. To achieve the objective of this project, all the required information for this project has been collected through different data collection such as interview, questioner and observation.

In this project, we apply object oriented approach to express the overall frame work of the system. The project has been implemented using now a day popular web technologies which are PHP for programming server side, MYSQL for data management, java script for validating form contents, HTML&CSS for formatting the interface.

Chapter One. Introduction

1.1 Background

Mekdela amba university (MAU) is one of the fourth generation universities in Ethiopia. it was established in 2015, the University has steadily grown and will be the one of the top education institutions in the country .

the Mekdela Amba University remains dedicated to contributing to the sustainable socio economic development of the nation through the provision of societal needs tailored education, undertaking problem based research and rendering relevant community services.

Mekdela Amba University (MAU) has five colleges and one school, focusing on Agriculture, Mining, Culture & Tourism, and Technology, offering various undergraduate/postgraduate programs

1.2 Statement of the Problem

As we observed from the current MAU registration system of students, Like most of other universities of our country the registration process of MAU is traditional and not comfortable to students and registrar workers, the reasons why we say this is as follows;

1. Once the registration date specified by the university and registration process starts, there is a lot of crudeness occurs in each windows of the registrar office of deferent departments of the university since there is one person who gives service for those students who need to register.
2. The students cannot get information easily during registration, or they need to attend notice board to get information. If the papers on the notice board tier or lost and such like things appears the probability of getting information is become low.

3. The other problem we see is the students sometimes submitted forms without filling them.
4. the current working condition of the MAU registrar system, if any student have compliant he/she need to contact with any of the registrar office member that person may or may not the exact person to replay the compliant.
5. And also in the existing registrar system there is no comfortable way for instructors to submitting grade for the registrar as well as students to see their result.

1.3 Objective of the Project

The general objective of this project is to automate the current manual system carried out by the university to register students using our online university registration system.

The specific objectives for our project are listed below:

- ➢ To reduce the time students take to register;
- ➢ To reduce the inconvenience students face during registration;
- ➢ To avoid the need to collect registration slips from the university; and

To help the University in reducing the labor needed to give services to students particularly during registration period.

1.4 Feasibility Study

For this project, we have considered two types of feasibility studies: technical and economic feasibility studies.

1.4.1 Technical Feasibility

Technical feasibility is about whether the technical resources needed to complete the project are met by the team members. For this project, the team, has all the necessary technical resources needed, whether it is technical knowledge, hardware or software needs, to accomplish the project we are undertaking. Therefore, our project is technically feasible to carry out.

1.4.2 Economic Feasibility

The newly developing system is economically feasible since there is no much cost to spend, Compared to how much benefits we get after developing the system. The expense of the newly developing system includes, costs for buying the different hardware and software which helps to reach to the desired solution of the problem. When we compare the current expense with the newly developing system, the existing system spends a lot of money for materials, like for papers, printer colors, copies, for notices etc. in each year and semesters. The new system also uses this materials but the quantity is not the same as the existing so the new system is beneficial Technical feasibility:

The newly developing system technically feasible, since it uses materials which are hardware software and computer technology which is available in the market and the system can be develop with our level of knowledge. Operational Feasibility To determine the operational feasibility of the system the team should take into

consideration the awareness level of the users. This system is operational feasible since the users are familiar with the technologies and hence there is no need to give

more training to use system. day for 24 hours. Schedule feasibility Concerning the project scheduling, it will be bound by proper timing so it must be delivered within the time bound given over the table

1.5 Significance of the Project

This project has the following significances:

- ➢ It reduces the time it takes students to register.

It allows students to register at anytime and anywhere.

- ➢ It allows the University to keep students information in a well-documented way.
- ➢ It reduces the time the University spends during registration time.
  It allows the University to attract more prospective university students because it offers online registration that most universities do not.
- ➢ Overall, it makes registration convenient for all students of the University.

1.6 Beneficiaries of the Project

The main beneficiaries of this project are the students and the University. ➢

Students

- It helps students to register without the need for any intervention.
- The project will reduce the time it used to take for the student to register.
- The students will have 24/7 service to register.

➢ The University

- It reduces paper (manual) work.
- It reduces the crowd seen at the registrar during registration.

1.7 Methodology

1.7.1 System Development Methodologies

System development methodology is used as a roadmap for any software project. It is used to structure, plan, and control the process development of a project. It is an important part of software engineering. In fact, we will be using one methodology for our project- a spiral method, which is a type of evolutionary process model.

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Spiral model is an evolutionary software process model that couples the iterative nature of prototyping with the controlled and systematic aspects of the waterfall model. It provides the potential for rapid development of increasingly more complete versions of the software. We chose this methodology because it is a combination of iterative model and waterfall model. Specifically, we chose spiral methodology for the following reasons:

- ➢ It offers iterative capability, which we are going to use in future increment of the project.
- ➢ Since this model is partly based on waterfall model, it allows us to follow linear process at first iteration.

We can add additional functionality or changes that can be done at a later stage.

- ➢ It will help us in developing the software fast.
- ➢ If any change occurs in the requirement of the project at later phases, those changes can be incorporated using this model.

1.8 Development Environment/programming tools and other tools

For our project, we will be using the following tools:

Windows 11 is our choice of operating system on which to develop our project.

- ➢ For the front-end part of our web-based system, we will use HTML, CSS, JavaScript and bootstrap as front-end framework.
- ➢ We will use PHP as back-end language.
- ➢ For database, we will use MySQL.

- ➢ For text-editor, we will use Visual studio Code.
- ➢ For browser, we will use Google Chrome We will use Apache as our server.
- ➢ We will be using Microsoft project and Microsoft Visio to draw different charts that are going to be used in our project.

We will use the following hardware tools for our project:

- ➢ Laptops
- ➢ USB Flash drive

1.9 Scope and Limitations of the Project

1.9.1 Scope of the Project

The scope of this project is listed below:

Our project is done in the MAU and the system boundary includes the two campuses of the university and it is applied to register students from first year up to graduate, regular program students the reason for not including remedial students is somehow remedial student's registration is takes time and has process. The system also allows online submission of grade by the instructors and enables students to view their grade. **Limitations of the Project**

Some of the limitations of our project are:

- ➢ It does not allow students to add and drop for the time being.
- ➢ It does not include mobile apps and desktop applications.

1.10 Risks, Assumptions and Constraints

1.10.1 Risks

The risks we may face during this project are:

- ➢ We may not be able to finish the project according to our schedule because we are carrying out our classes side by side, and we have to study other courses, too.

- ➢ The requirement for the project may suddenly change if the University decides to change the way it carries the registration process, making it difficult for us to cope up with these sudden changes.
- ➢ If we get behind the project schedule, we may leave some features of the proposed system.

The scope of the project might become too large and result in failure

1.10.2 Assumptions

The assumptions that we are going to make for this project are:

- ➢ We are assuming that the requirement for this project will not change in a way that will have a considerable effect on our project.
- ➢ Another assumption we are making is that we will be able to finish the project within the schedule.

  Additionally, we are assuming that we will get the necessary information needed for our project from the Registrar Office and from students of the University.

1.10.3 Constraints

Some of the constraints of this project are:

- ➢ Since we will be carrying out the project side by side with attending class, we may have a limited time to carry out the project.
- ➢ Another constraint for this project is that our proposed system is web based and only works for Mekdela amba university

Chapter Two. Business Area Analysis and Requirement Definition

2.1 Introduction

This chapter mainly deals with the inner working of the system used by the Registrar Office and about the requirements this project must satisfy.

To get detailed and thorough information about the existing system, we have used our own observation in addition to carrying out interviews with personnel at the Registrar Office.

2.2 Business Area Analysis

2.2.1 Business Rule Identification Rule

1

- ➢ The person should be a student of the Mekdela amba University and should have an ID card. Rule 2
- ➢ To register for a course, the student should have taken and passed the prerequisite for the course. Rule 3
- ➢ Registration is carried out within a time limit specified by the Registrar Office.

Rule 4

The number of credit hours per semester the student is registering for should not exceed that of the number set by the University.

Rule 5

- ➢ If a student did not register within the set deadline for the registration, he/she can register during the late registration period, with penalty.

Rule 6

- ➢ For new students their original documents should be verified and a copy of their original documents should be submitted.

Rule 7

- ➢ New students should have a result above the entrance point set by the Ministry of Education. Rule 8

  In order to enroll, the courses the student is registering for the semester should be available.

2.2.2 Current System Vs Proposed System

Current System	Proposed System
➢ Students cannot register by themselves, which means there is the need for a registrar clerk as an intermediary for any student to register.	➢ Our proposed system will allow students to register all by themselves eliminating the need for another person in the registration process.
➢ Students can only register in-person during registration period.	✓ Since our proposed system will be an online system students will be able to register from anywhere, they are and they do not have to be present in-person.
➢ Because the registrar office only works during workdays and work hours, students cannot register outside the workdays and workhours.	✓ The system will be available for twenty-four hours and as such, students can register at any time they want within the registration period.
✓ The current system is not fully automated and therefore students' information is kept in papers.	✓ The proposed system is fully automated and because of that, all of students' information is kept in a database.

2.2.3 Players of the Existing System

Student

✓ To get information about the registration and make them to access it; ✓ To register for course online by just filling a forms and submit easily and To add their comments.

Staff

✓ To submit grades of students online; ✓ To see information

posted online and ✓ To add their comments.

Registrar

✓ To check registered student's list easily;

✓ To post announcements easily and pass message to students and ✓ View comments of users easily.

2.3 Requirement Definition

2.3.1 Functional Requirements

The online student registration system should be designed in order to Validate customer:-when the user inserts user name and password to log in to the system, the system will validate weather user name/password correct or not. If correct open the next page else the system displays message to the user to insert username and password again.

- **Provide a home page which lets the users to select student login, and administer login and staff member login.**
- **Provide registration form for users and let the user to create account (user name, password)**
- **Provide searching operation based on the user preference.**
- **Provide reports like credit, schedule, grade report and registered courses for the user.**
- **Let the instructors to submit grades of courses via grade submission form.**
- **Delete user: the system should let the system administrator to delete user who are not use the system for specified time length.**
- **View user information.**
- **Change password of the user.**

2.3.2 None Functional Requirements

- Speed: - The speed of the new system is much higher the time needed to get any required information as quickly as possible.

- Cost: - The cost is very less as compared with its use.
- Performance and Efficiency: - It has improved performance the system must respond to user requests as fast as possible and fully operational every day for 24 hours.
- Energy: - the new system reduces energy because it decreases paper work, saves energy from moving here and there.

Error handling: - has error controlling mechanism reducing error rates due to validation.

- Security: - the system is more secured since it needs to enter user name and password.
- User interface: - the application should have a user interface which is simple to use and navigate through each page.

2.4 Requirement Gathering

Requirements gathering are one of the most essential parts of any project. It adds value to a project on multiple levels because it is where all the requirements for the project are gathered.

It is an exploratory process that involves researching and documenting the project's exact requirements from start to finish.

In our project, to identify the requirements expected from all stakeholders, including students and registrar staff, we have used different techniques.

2.4.1 Requirement Gathering Techniques

When it comes to gathering requirements, the best way to do so differs among projects. Some requirement gathering tools and templates may work better for some projects but not for others. In fact, we have used different techniques that we think will help us in extracting as much information as we can get.

Below are the requirement gathering techniques we used in this project:

- ➢ Interview
- ➢ Brainstorming
- ➢ Observation
- ➢ Document Analysis
- ➢ Questionnaire

2.5 Method of Communication

Communication is a critical factor in project management. There are instances where projects have failed because of miscommunication and communication gaps. A project that is effectively communicated among team members will proceed smoothly and without much disagreement between members. This will eventually lead to achieving the project's goal within the deadline.

Chapter Three: System Analysis and Design

3.1 Introduction

System Analysis and Design is an object-oriented approach that elaborates the overall design and structure of any system. System Analysis and Design helps in having a clear image of a system.

This chapter is about the design of the system, and it includes many structural designs in different forms such as activity diagram, use case diagram, and sequence diagram. It is the most important part of any software project because if a system is well analyzed and designed before it is implemented, then the remaining stages will be relatively easy to do. Therefore, we have spent a lot of time and effort to come up with a good design after analyzing the overall system. This chapter includes:

- Use case Diagram
- Activity Diagram

- Sequence Diagram
- Class Diagram

These diagrams and descriptions help us to map out what the system should include, how it should work, and what it will look like after it is implemented.

3.2 System Use Case

3.2.1 Actor Description

- **Existing student** is a person who has the right to register, view his or her grade and other things on the system.
- **New student** is a person who has the right to create account and register to the system.
- **Instructor** is a person who has the right to upload course result, change course result and other things on the system.

System admin is a person who has the right to create account for the Registrar Officer, instructors, department staff and have the mandate to delete their account.

- **Department** is a person who has the responsibility for uploading courses.

3.2.2 Use Case Diagram Design

Use case diagram is used to show the interaction between different actors and use cases. In the above diagram, six actors interact with different use cases. All the actors except new students interact with login and change password use case; on the other hand, system admin interacts with use cases that are related to creating accounts for different users. Instructor interacts with upload course result, view course result, while students both new and existing register, and verify payment. The use cases related to registrar officers are edit calendar, approve new student and view grade of students.

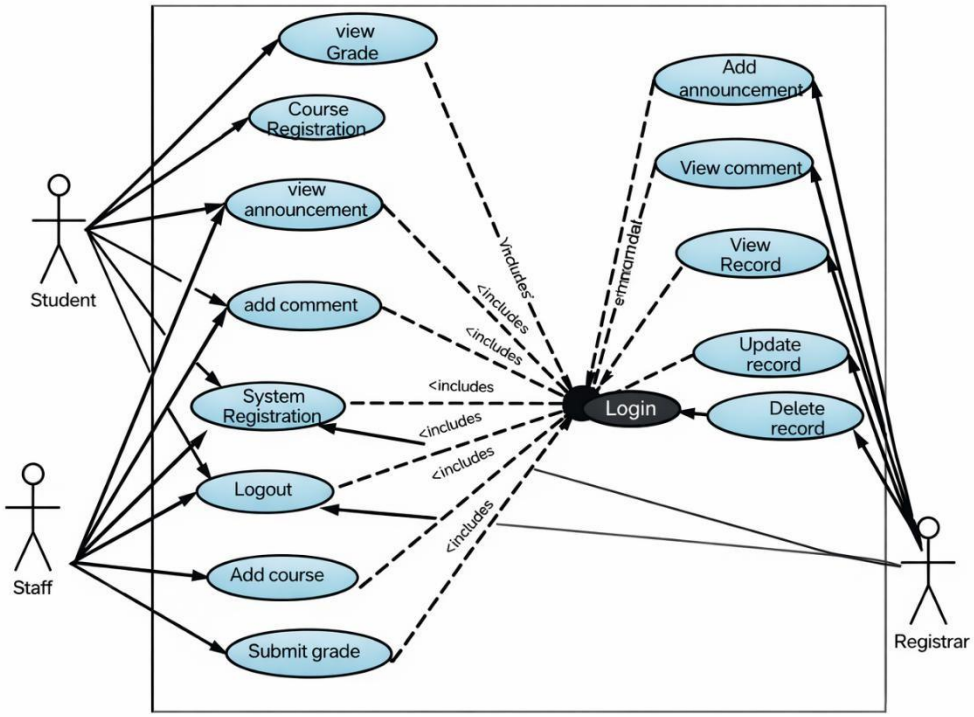


Figure 3.1 Use case diagram

3.3 Activity Diagram Design

An activity diagram visually presents a series of actions or flow of control in a system. They can also describe the steps in a use case diagram. Each activity diagram shows the actions taken to achieve a specific activity.

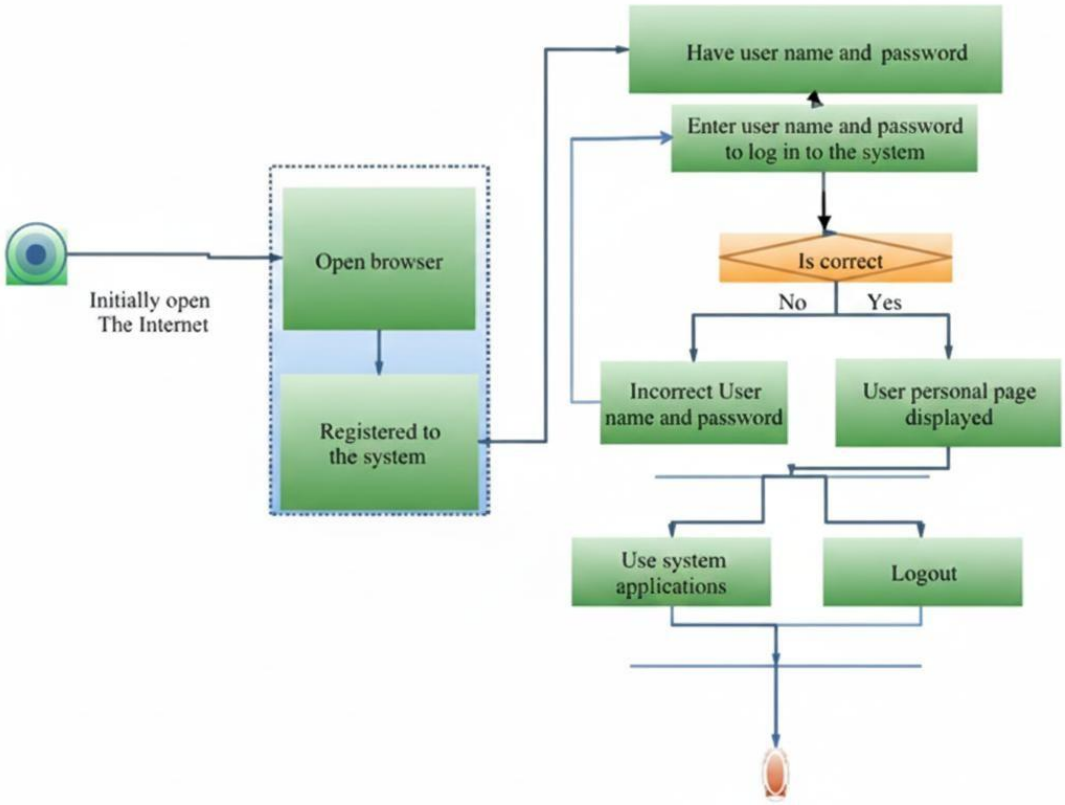


Figure 3.2 Activity diagram

3.4 Sequence Diagram Design

Sequence Diagrams are interaction diagrams that detail how operations are carried out. They capture the interaction between objects. Sequence Diagrams are time focused and they show the order of the interaction visually by using the vertical axis of the diagram to represent what messages are sent and when.

Every sequence diagram we designed for our project shows the interaction between an actor and different pages of the system, and the database where all data is stored.

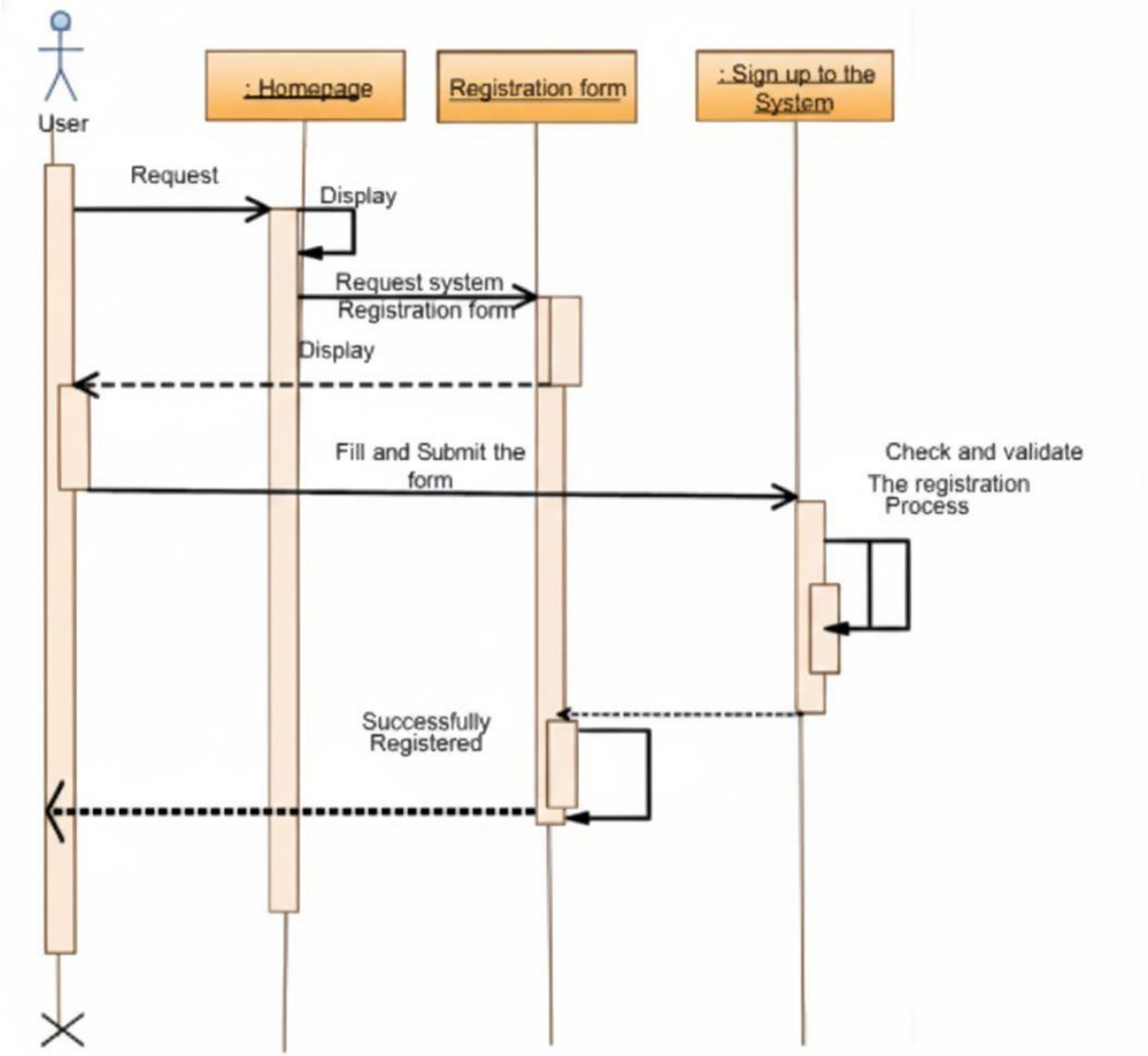


Figure 3.3 Sequence diagram sign up to the system.

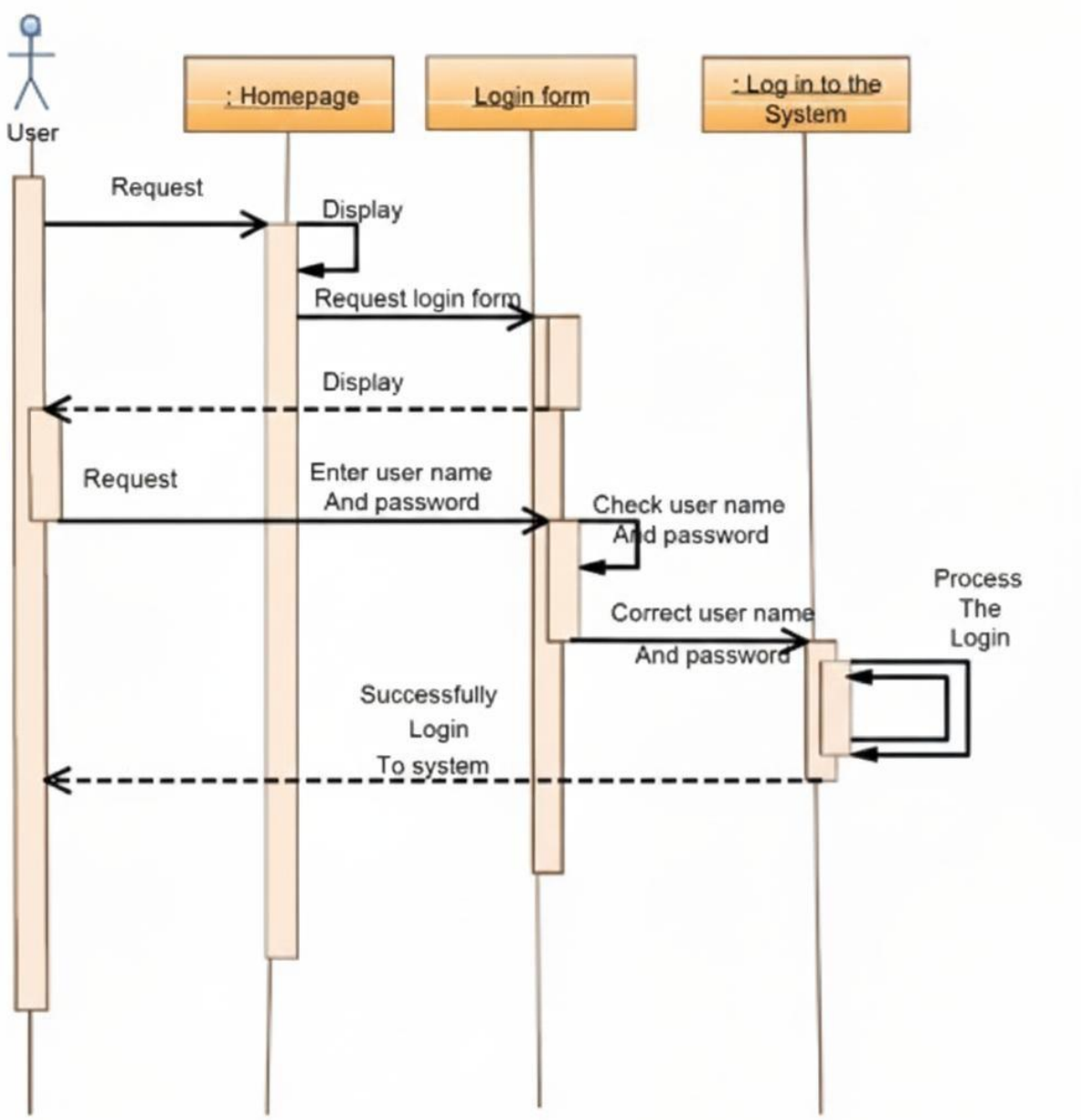


Figure 3.4 Sequence diagram for login to the system.

3.5 Class Diagram Design

A class diagram is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among objects. For this project, there are eight classes: System Admin, Account, Student, Department, Instructor, Department staff, Course, and registrar officers. Each of the classes has different methods.

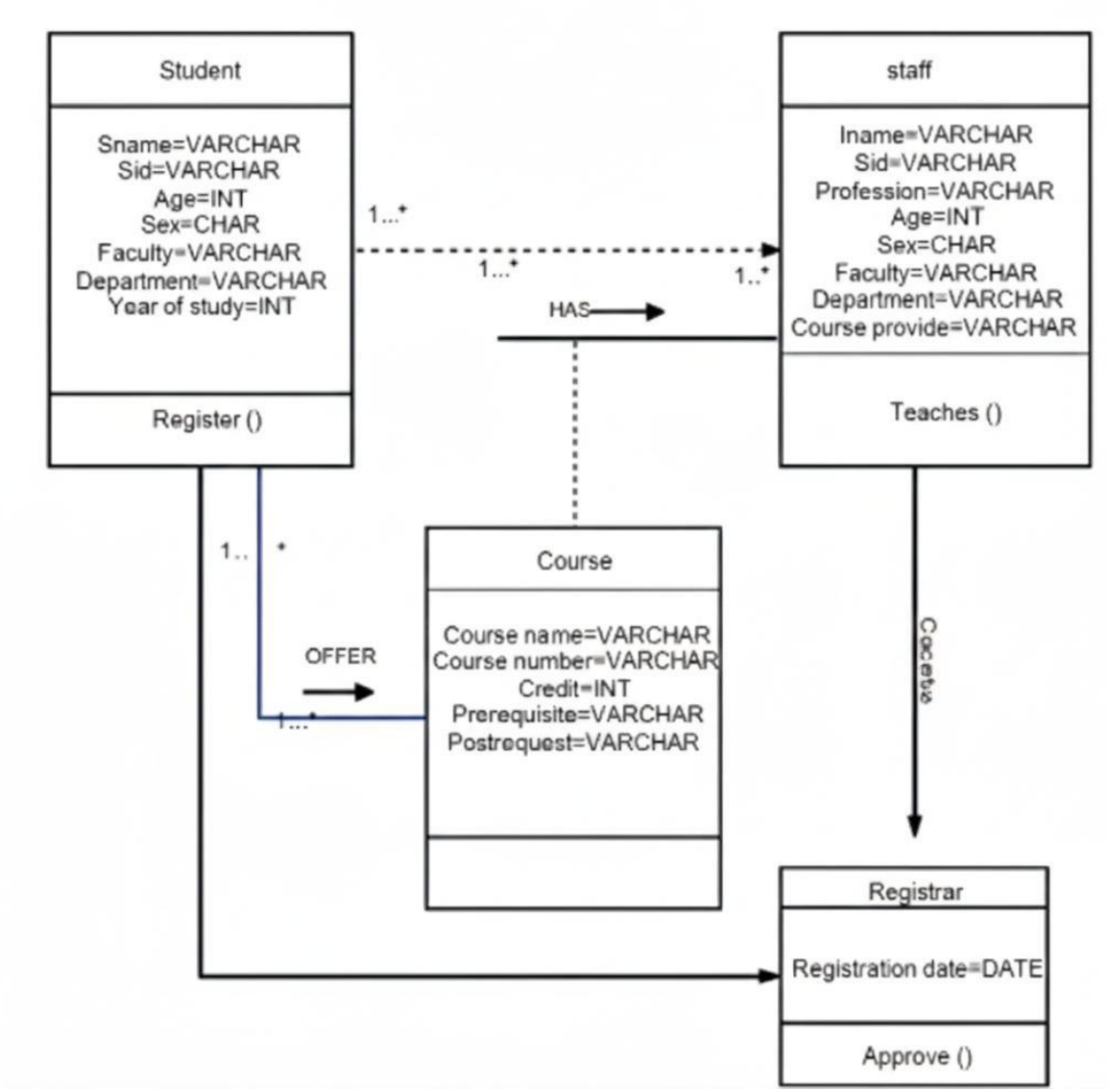


Figure 3.5 class diagram

3.6 Database Modeling

Data modeling is the process of creating a data model for the data to be stored in a database. A data model describes information in a systematic way that allows it to be stored and retrieved efficiently in a Relational Database System. The goal is to illustrate the types of data used and stored within the system, the relationships among these data types, the ways the data can be grouped and organized, and its formats and attributes.

3.6.1 Database Schema

We have the following database tables in our system that is used to store user information.

- Student
- Course
- Department
- Instructor
- Registrar officer
- System Admin
- Account

3.6.2 What is an ER Diagram?

An Entity–Relationship (ER) diagram is a visual representation of the data structure in a database. It shows:

- **Entities → the main objects or concepts (e.g., *Student*, *Course*, *Teacher*).**
- **Attributes → properties of entities (e.g., *Student Name*, *Course ID*).**
- **Relationships → how entities are connected (e.g., *Student enrolls in Course*).**

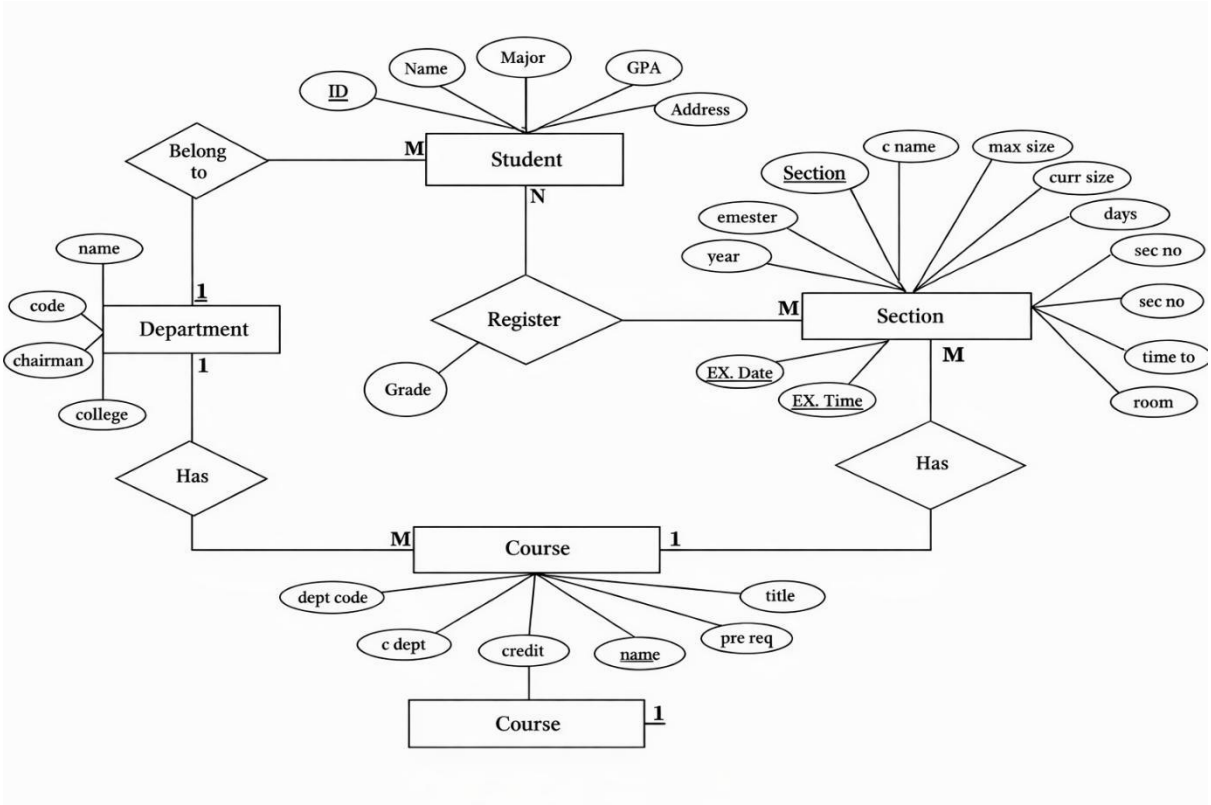


Figure 3.6 ER diagram

3.6.2 Normalization

Normalization is a database design technique that reduces data redundancy and eliminates undesirable characteristics like Insertion, Update and Deletion Anomalies. Normalization rules divide larger tables into smaller tables and links them using relationships. For this project, we have tried to normalize up to third normal form.

3.7. Implementation

This is the phase where actual production of the proposed project is done according to the functional requirements outlined in Chapter Two and the system design done in Chapter Three.

In relation to the software process model used in this project, which is spiral model, this phase maps into the construction and deployment phase of spiral model.

Upon the start of this phase, the tools and environment needed for the implementation is prepared and ready. On the completion of this phase, the system will be fully implemented and ready to be used. Furthermore, we are assuming that the system will be installed and end users are introduced to the system and have the knowhow of the system, since we do not have the permission to access the University's server to actually install the system.

3.4.1 Technologies

Once our website is up and running, end users like students, instructors and registrar employees will want to access the website. To do this, they must provide an address to the URL (Uniform Resource Locator) in the web browser, the browser in return requests the needed file via HTTP. When the request reaches the web server, the HTTP server accepts the request, finds the requested document, and sends it back to the browser through HTTP. If the server doesn't find the requested file, it returns a 404 response instead which indicates that the server cannot find the requested resource.

For our project, we are using Apache HTTP Server to locally host our website and all the related files. The purpose of a web server is to present the content of the website to the user's browser. These contents can be web pages or media files such as images, videos or plain text files. These files each have their own file extension, such as a file with "CSS" extension indicates that the file is cascading style sheet and "php" extension indicates that the file is a php file and such files will be processed by the web server and render the necessary html file and send it to the web browser.

Our website is a dynamic website which means that the end users can interact with it by filling different forms when there is the need to store information that the users entered. So, a database is needed, and for our project we are using MySQL database server. We will be using MySQL database to store students', instructors', and other users' data persistently. MySQL database is a popular database that works well with Apache Server.

For our project we will be using XAMPP server which consists mainly of the Apache HTTP Server, XAMPP will help in developing our website in a fast pace since the time needed to configure the web server, database server and php engine separately is reduced significantly. Using XAMPP sever ,we are able to mimic a real live server in

a local host hence making it easy to transition from local to live server during deployment phase.

3.4.2 Apache Server

Apache HTTP Server is a free and open-source web server that delivers web content through the internet. The Apache HTTP Server ("httpd") was launched in 1995 and it has been the most popular web server on the Internet since April 1996. Currently, it is on its 2.4.48 version. Its purpose is to establish a connection between a physical server and the browsers of website visitors (Firefox, Google Chrome, Safari, etc.) while delivering files back and forth between them. Apache servers process files written in different program languages such as Python, Java, and in our case PHP and others which the web browser cannot understand.

Then it turns them to static HTML files and serves these files in the browser for web users. Features of Apache Web Server:

- Handling of static files
- Loadable dynamic modules
- Compatible with IPv6
- Supports HTTP/2
- FTP connections
- Gzip compression and decompression Bandwidth throttling
- Compatible with different languages like php, python and Perl…etc.
- Load balancing
- Session tracking
- URL rewriting
- Geo location based on IP address

Initially Apache server was developed for UNIX but throughout the years it has become compatible with Windows operating system as well. But usually, it is used

with Linux operating system to form the LAMP stack which includes Linux OS, Apache server, MySQL database, and PHP as the backend scripting language.

3.4.3 PHP

PHP (PHP: Hypertext Preprocessor) is an open-source, interpreted, and object-oriented scripting language that can be executed at the server-side which makes it well suited for web development. PHP code is processed on a web server by a PHP interpreter implemented as a module or a daemon. On a web server, the result of the interpreted PHP code which may be any type of data, such as generated HTML would form the whole or part of an HTTP response.

PHP is used to manage dynamic content, databases, and session tracking. PHP can handle forms, gather data from files, save data to a file, send data, and return data to the user. It can add, delete, modify elements within the database. It can access cookies variables and set cookies. Using PHP, we can restrict users to access some pages of your website.

Features of PHP:

- Database support
- Error reporting
- Platform independent

Security

- Loosely typed language

Since PHP is platform independent it can work on any OS and web server.

3.4.4 Integrating Apache server and PHP

Upon a request from the user web browser, a request is sent to the web server (Apache), then this request is sent to the Apache web server through the internet. Once Apache receives the request, it analyzes the file extension of the requested file and decides whether to send the file to the PHP interpreter.
If the file has php extension, then it will be sent to the PHP interpreter. The PHP interpreter then executes the php file. During execution of this file, PHP may interact with the database to fetch information. After PHP finishes executing the file it sends the output back to Apache server. Finally, Apache receives the output from PHP through the internet and sends back the response to the web browser where the response is displayed to the computer.

3.4.5 MySQL Server Database and Connectivity

MySQL is an open-source SQL relational database management system that's developed and supported by Oracle. It is one of the most popular languages for accessing and managing the records in the table. It is based on a client-server model. The core of MySQL is MySQL server, which handles all of the database instructions (or commands). MySQL server is available as a separate program for use in a client-server networked environment and as a library that can be embedded (or linked) into separate applications.

MySQL operates along with several utility programs which support the administration of MySQL databases. Commands are sent to MySQL Server via the MySQL client, which is installed on a computer.
MySQL runs on virtually all platforms, including Linux, UNIX and Windows and is supported by different web servers including Apache Server and IIS.

Features of MySQL Server database:

- Scalable
- Speed
- Memory efficiency

- High performance
- Secure
- Stored procedures, triggers, allows roll-back

Multiple storage engines like Inno DB (default since MySQL version 5.5), MyISAM, Merge, Memory (heap), Federated, Archive, CSV, Black hole, NDB Cluster.

- To connect to MySQL database using PHP, we can use MySQLi extension (MySQL improved) or PDO (PHP Data Objects), for our project we have used both PDO extension and MySQLi extension.

MySQLi supports both procedural and object-oriented programming.

PHP Data Objects (PDO) extension defines a lightweight, consistent interface for accessing databases in PHP. PDO_MYSQL is the driver that is used to implement the PHP Data Objects (PDO) interface to enable access from PHP to MySQL databases.

To use this driver in our project, we must enable the extension in the php.ini file.

3.4.6 Sessions

A session is a way to store information (in variables) to be used across multiple pages. By default, session variables last until the user closes the browser. Session identifiers or SID is a unique number which is used to identify every user in a session-based environment. These session IDs are randomly generated by the PHP engine. If we want to replace the systemgenerated session id with our own, we can supply it to the first argument of the session id function.

3.4.7 MVC

The Model-View-Controller (MVC) framework is an architectural pattern that separates an application into three main logical components. Model, View, and Controller. Each architectural component is built to handle specific development aspects of an application.

MVC separates the business logic and presentation layer from each other.

- Model: It includes all the data and its related logic
- View: Present data to the user or handles user interaction
- Controller: An interface between Model and View components

In our project, we have built our own MVC framework to implement the MVC pattern.

3.4.8 Testing

Software testing is the process of evaluating and verifying to check that a software product or application does what it is supposed to do. It is Important because if there are any bugs or errors in the software, it can be identified early and can be solved before delivery of the software product. Properly tested software product ensures reliability, security and high performance which further results in time saving, cost effectiveness and customer satisfaction.

For our project, we have used continuous testing as we kept adding functionality and features. Continuous testing is testing each build as it becomes available. This will allow us to find errors early on the development stage and fix these errors in time.

In our project, we have carried out the following testing methods:

- Unit Testing
- Integration Testing
- System Testing

Chapter Four: Conclusion and Recommendations

4.1 Conclusion

Many universities and educational institutions have a wide range of courses, students and faculty. Combine this with the need to conserve resources, and it makes sense to computerize the registration process. Registering for classes online reduces paperwork, personnel, and costs.

Our project's main objective was to allow students to register by themselves with minimal intervention from registrar workers. And throughout this project we have implemented this proposed system. In addition to registering students it allows instructors to enter students grade, and view grades for the course they have taught; and the department uploads the courses students must take for a specific year and semester; while registrar officers are able to approve new students, suspend students, view report for a specific year and semester. Furthermore, the system administrator is able to create accounts for employees that use the system like instructors, department officers, and registrar officers and as a report he/she will be able to see the details of the account that they have created and how many accounts they have created. Even though we have implemented most of what we have set out to do, there still remains part of the system that is not implemented and we are leaving this as groundwork for future implementation.

4.2 Recommendations

This system makes the registration process very easy for new students and accessible to qualified students. In addition, it integrates grade viewing by the student, managing students by the registrar, and uploading course result by the instructor making it easy for management and for the university's decision makers to improve their decision-making process by seeing reports generated by the system. So, we recommend that this system is implemented in real environment to allow every actor involved and the university get the full benefit of the system.

5. References

The Apache Software Foundation. (2021, August 19). *Documentation* . Retrieved from Apache HTTP Server Project: https://httpd.apache.org/

Boehm, B. (1988). A Spiral Model for Software Development and Enhancement. *Computer, 21*, 61-72.

Bootstrap. (2021, July 25). *Documentation* . Retrieved from Bootstrap: https://getbootstrap.com/docs/4.0/getting-started/introduction/

Hernandez, J. (2019, May 8). A*pache Webserver Introduction*. Retrieved from Sumo Logic: https://www.sumologic.com/blog/apache-web-server-introduction/ MDN.

(2021, Sepetember 20). *MVC*. Retrieved from MDN Web Docs: https://developer.mozilla.org/en-US/

phptutorial. (2021, August 14). *Connecting to MySQL*. Retrieved from phptutorial: https://www.phptutorial.net/php-pdo/pdo-connecting-to-mysql/

Pressman, R. S. (2010). *Software Enginneering: A Practitioner's Approach* (7th ed.). New York: The McGraw-Hill Companies.

Rob, P., & Coronel, C. (2009). *Database Systems: Design, Implementation, and Management* (8th ed.). Boston: Thompson Course Technology.

Satzinger, J. W., Jackson, R. B., & Burd, S. D. (2012). *System Analysis and Design in a Changing World* (6th ed.). Boston: Cengage Learning.

Sommerville, I. (2011). *Software Engineering* (9th ed.). Boston: Pearson Education.

Tatroe , K., & MacIntyre, P. (2020). *Programming PHP* (4th ed.). Sebastopol: O'Reilly Media.

Valacich, J. S., & George, J. F. (2017). *Modern Systems Analysis and Design* (8th ed.). Boston: Pearson Education.

w3schools. (2021, February 23). *php mysql intro*. Retrieved from w3schools: https://www.w3schools.com/php/php_mysql_intro.asp